STAT 1210: Assignment 3 Solutions

Part III: Foundations of Inference

Fall 2026 Semester

Question	**Marks**
Question 1	10
Question 2	20
Question 3	20
Question 4	25
Question 5	25
Total	**100**

Question 1 [10 marks]

1. Proportions and association **[5 marks]**

Among dogs receiving at least 60 minutes of activity per day, the sample proportion classified as obese is

$$\widehat{P}(\text{obese} \mid \text{at least 60 minutes}) = \frac{18}{150} = 0.120$$

Therefore, 12.0% of the dogs in the higher-activity group were classified as obese. Among dogs receiving less than 60 minutes of activity per day, the sample proportion classified as obese is

$$\widehat{P}(\text{obese} \mid \text{less than 60 minutes}) = \frac{126}{450} = 0.280$$

Therefore, 28.0% of the dogs in the lower-activity group were classified as obese. The difference between the sample proportions is

$$0.280 - 0.120 = 0.160$$

The obesity percentage was therefore 16 percentage points higher in the lower-activity group. Equivalently, the observed obesity proportion in the lower-activity group was

$$\frac{0.280}{0.120} \approx 2.333$$

times the proportion in the higher-activity group. The sample shows an association between lower reported daily activity and a higher obesity proportion.

Suggested allocation: 1 mark for each correct conditional proportion, 1 mark for converting the results to percentages, and 2 marks for an accurate contextual comparison.

2. Study design and limitations **[5 marks]**

This is an **observational study**. The researchers recorded activity and obesity status but did not assign dogs to activity groups.

Random sampling was not used. The five clinics volunteered, and participation was limited to owners who visited those clinics during the two-week recruitment period and agreed to participate. Random assignment was also not used because owners and dogs were not assigned by chance to activity levels.

One plausible confounding variable is diet. Dogs receiving higher-calorie diets may be more likely to become obese and may also be less active. Other reasonable examples include age, breed, existing illness, medication use and owner lifestyle.

One possible source of selection bias is that dogs visiting the volunteer clinics may differ from adult dogs that did not receive veterinary care during the recruitment period. Voluntary participation by owners could also create voluntary-response or nonresponse bias if participating owners differ systematically from those who decline.

Because there was no random assignment, the observed association could be explained partly or entirely by confounding. The study therefore cannot establish that low activity *causes* obesity. Because the dogs were not randomly sampled from all adult dogs in Prince Edward Island, the results also cannot be confidently generalized to that full population.

Suggested allocation: 1 mark for identifying an observational study, 1 mark for the random-sampling assessment, 1 mark for the random-assignment assessment, 1 mark for a plausible confounder and source of selection bias, and 1 mark for the causal and generalizability conclusions.

Question 2 [20 marks]

1. Parameter, statistic, centre and spread **[5 marks]**

The population parameter is

$$p = 0.42$$

the population proportion of tagged juvenile Atlantic salmon that would survive the migration route under the stated conditions. The sample statistic is $\hat{p}$, the proportion of the 400 sampled salmon that survive. The mean of the sampling distribution is

$$\mu_{\hat{p}} = p = 0.42$$

So across many random samples of 400 salmon, the average value of $\hat{p}$ would be 0.42. This shows that $\hat{p}$ is an unbiased estimator of p. The standard deviation is

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{(0.42)(0.58)}{400}} = \sqrt{\frac{0.2436}{400}} = \sqrt{0.000609} \approx 0.024678$$

Across repeated random samples of 400 salmon, the sample survival proportion typically differs from 0.42 by approximately 0.0247, or 2.47 percentage points.

Suggested allocation: 1 mark for the parameter and statistic, 1 mark for the centre, 2 marks for the standard-deviation calculation, and 1 mark for the interpretations.

2. Conditions and approximate sampling distribution **[5 marks]**

The expected number of surviving salmon is

$$np = 400(0.42) = 168$$

The expected number of salmon that do not survive is

$$n(1-p) = 400(0.58) = 232$$

Both expected counts are greater than 10, so the success–failure condition is satisfied. The problem also states that the outcomes may be treated as independent and that the population is sufficiently large relative to the sample. Therefore, the sampling distribution of $\hat{p}$ is approximately Normal, with

$$\mu_{\hat{p}} = 0.42 \qquad \text{and} \qquad \sigma_{\hat{p}} \approx 0.024678$$

In notation that explicitly identifies the standard deviation

$$\hat{p} \text{ is approximately Normal with mean } 0.42 \text{ and standard deviation } 0.024678$$

Suggested allocation: 1 mark for each expected count, 1 mark for confirming the condition, and 2 marks for correctly stating the approximate distribution.

3. Calculating $P(\hat{p} \leq 0.38)$ **[5 marks]**

The standardized value is

$$z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} = \frac{0.38 - 0.42}{0.024678} \approx -1.621$$

Using $z = -1.62$ in the Standard Normal table gives

$$P(\hat{p} \leq 0.38) \approx P(Z \leq -1.62) = 1 - P(Z \leq 1.62) = 1 - 0.9474 = \boxed{0.0526}$$

Thus, if the population survival probability is 0.42, approximately 5.26% of random samples of 400 salmon would have a sample survival proportion of 0.38 or lower.

Suggested allocation: 2 marks for standardization, 2 marks for using the correct Normal-table area, and 1 mark for the contextual interpretation.

4. Calculating $P(0.40 \leq \hat{p} \leq 0.46)$ **[5 marks]**

The lower standardized boundary is

$$z_L = \frac{0.40 - 0.42}{0.024678} \approx -0.810$$

The upper standardized boundary is

$$z_U = \frac{0.46 - 0.42}{0.024678} \approx 1.621$$

Using the rounded table values $z_L = -0.81$ and $z_U = 1.62$ gives

$$P(0.40 \leq \hat{p} \leq 0.46) \approx P(-0.81 \leq Z \leq 1.62) = P(Z \leq 1.62) - P(Z \leq -0.81) = 0.9474 - 0.2090 = \boxed{0.7384}$$

Using unrounded standardized values gives approximately 0.7386. Either answer is acceptable when the calculation is shown consistently.

Suggested allocation: 1 mark for each standardized boundary, 2 marks for subtracting the correct cumulative probabilities, and 1 mark for the final probability.

Question 3 [20 marks]

1. Parameter and point estimate **[5 marks]**

Let μ denote the population mean recovery time for all adult dogs in the hospital network who underwent the routine procedure during the study period. The point estimate of μ is the sample mean

$$\bar{x} = 47.8 \text{ minutes}$$

Suggested allocation: 3 marks for defining the population mean with the correct population and measurement, and 2 marks for identifying the sample mean as the point estimate.

2. Conditions for a one-sample z interval **[5 marks]**

- **Random sampling:** The problem states that a simple random sample of 64 dogs was selected.
- **Independence:** A simple random sample supports independence. The 10% condition provides additional support when sampling without replacement.
- 10% **condition:** The network treated more than 1000 dogs, and

  $$64 < 0.10(1000) = 100$$

  Therefore, the sample includes less than 10% of the population.
- **Sampling-distribution shape:** The population shape and raw data are not provided, so skewness and outliers cannot be checked directly. However, $n = 64$ is sufficiently large for the Central Limit Theorem to support an approximately Normal sampling distribution of $\bar{x}$, provided the population has finite variability and is not dominated by extreme observations.
- **Known population standard deviation:** The problem states that the population standard deviation is treated as known at $\sigma = 12$ minutes, so a z interval is appropriate under the stated model.

Suggested allocation: 1 mark for each condition or assumption assessed correctly.

3. The 95% confidence interval **[5 marks]** The standard error is

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{12}{\sqrt{64}} = \frac{12}{8} = 1.5 \text{ minutes}$$

For a 95% confidence interval

$$z^* = 1.960$$

The margin of error is

$$m = z^* SE = 1.960(1.5) = 2.940 \text{ minutes}$$

The confidence interval is

$$\bar{x} \pm m = 47.8 \pm 2.94 = (47.8 - 2.94,\ 47.8 + 2.94) = \boxed{(44.86,\ 50.74) \text{ minutes}}$$

We are 95% confident that the population mean recovery time for adult dogs undergoing this procedure in the hospital network during the study period is between 44.86 and 50.74 minutes. The confidence level describes the long-run success rate of the interval procedure, not a probability attached to the fixed parameter after this interval has been calculated.

Suggested allocation: 1 mark for the standard error, 1 mark for the critical value and margin of error, 1 mark for the endpoints, and 2 marks for an accurate contextual interpretation.

4. The 99% confidence interval and comparison **[5 marks]**

For a 99% confidence interval

$$z^* = 2.576$$

The margin of error is

$$m_{99} = 2.576(1.5) = 3.864 \text{ minutes}$$

Therefore

$$47.8 \pm 3.864 = (47.8 - 3.864,\ 47.8 + 3.864) = \boxed{(43.936,\ 51.664) \text{ minutes}}$$

The width of the 95% interval is

$$2(2.940) = 5.880 \text{ minutes}$$

The width of the 99% interval is

$$2(3.864) = 7.728 \text{ minutes}$$

The 99% interval is wider because a higher confidence level requires a larger critical value. With the sample size and population standard deviation unchanged, the larger critical value produces a larger margin of error.

Suggested allocation: 2 marks for the margin of error and endpoints, 1 mark for each width, and 1 mark for explaining the confidence–precision trade-off.

Question 4 [25 marks]

1. Parameter and hypotheses **[5 marks]**

Let μ denote the population mean resting respiratory rate during recovery for adult harbour seals treated with the revised protocol. The hypotheses are

$$H_0 : \mu = 18$$

$$H_a : \mu \neq 18$$

The alternative is two-sided because respiratory rates below or above 18 breaths per minute may both be clinically important.

Suggested allocation: 1 mark for the parameter, 1 mark for H_0, 1 mark for H_a, and 2 marks for explaining the two-sided alternative.

2. Sample mean, standard error and test statistic **[5 marks]**

The sum of the 25 measurements is

$$\sum_{i=1}^{25} x_i = 503$$

The sample mean is

$$\bar{x} = \frac{\sum x_i}{n} = \frac{503}{25} = 20.12 \text{ breaths per minute}$$

Under the null model, the standard error is

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{4}{\sqrt{25}} = \frac{4}{5} = 0.8 \text{ breaths per minute}$$

The test statistic is

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{20.12 - 18}{0.8} = \frac{2.12}{0.8} = \boxed{2.65}$$

The positive sign indicates that the observed sample mean is above the null value. Its magnitude indicates that the sample mean is 2.65 standard errors above 18 breaths per minute.

Suggested allocation: 2 marks for the sample mean, 1 mark for the standard error, 1 mark for the test statistic, and 1 mark for its interpretation.

3. P-value, decision and conclusion **[5 marks]**

Because the alternative hypothesis is two-sided, equally extreme results in both tails must be included. The Standard Normal table gives

$$P(Z \leq 2.65) = 0.9960$$

So the upper-tail area is

$$P(Z \geq 2.65) = 1 - 0.9960 = 0.0040$$

The two-sided P-value is

$$P\text{-value} = 2P(Z \geq 2.65) = 2(0.0040) = \boxed{0.0080}$$

Because

$$0.0080 < 0.05$$

we reject H_0. At the 5% significance level, the data provide statistically significant evidence that the population mean resting respiratory rate under the revised protocol differs from 18 breaths per minute. The sample result suggests that the mean is higher than 18, although the test was designed to detect differences in either direction.

Suggested allocation: 2 marks for the two-sided P-value, 1 mark for comparison with α, 1 mark for the decision, and 1 mark for the contextual conclusion.

4. Type I and Type II errors **[5 marks]**

A **Type I error** would occur if the team concluded that the population mean respiratory rate under the revised protocol differs from 18 breaths per minute when the true population mean is actually 18.

A **Type II error** would occur if the team failed to detect a difference from 18 breaths per minute when the revised protocol actually produces a population mean that is different from 18.

From an animal-safety perspective, a Type II error may be more serious if it allows a genuinely unsafe protocol to remain in use. A Type I error could also have costs, such as rejecting a safe and useful protocol. Full credit should be given for either choice when the clinical consequences are explained clearly.

Suggested allocation: 2 marks for each error stated in context and 1 mark for a justified comparison of their consequences.

5. Confidence interval and clinical importance **[5 marks]**

The 95% margin of error is

$$m = 1.960\left(\frac{4}{\sqrt{25}}\right) = 1.960(0.8) = 1.568$$

The interval is

$$\bar{x} \pm m = 20.12 \pm 1.568 = (20.12 - 1.568,\ 20.12 + 1.568) = \boxed{(18.552,\ 21.688) \text{ breaths per minute}}$$

The null value 18 does not lie in this interval. This agrees with rejection of the two-sided null hypothesis at $\alpha = 0.05$ because a two-sided 5% test corresponds to a 95% confidence interval under the same model.

Statistical significance alone does not establish clinical importance. The estimated increase and the interval endpoints must be compared with clinically meaningful respiratory-rate thresholds and animal-health outcomes.

Suggested allocation: 1 mark for the margin of error, 2 marks for the endpoints, 1 mark for the connection to the test, and 1 mark for distinguishing statistical from clinical importance.

Question 5 [25 marks]

1. Test statistic, P-value and decision **[5 marks]**

Let μ denote the population mean of portable-meter reading minus reference-laboratory reading for canine blood samples in the target setting. The standard error is

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{2.4}{\sqrt{400}} = \frac{2.4}{20} = 0.12 \text{ mg/dL}$$

The test statistic is

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{0.30 - 0}{0.12} = \boxed{2.50}$$

The two-sided P-value is

$$P\text{-value} = 2P(Z \geq 2.50) = 2\{1 - P(Z \leq 2.50)\} = 2(1 - 0.9938) = \boxed{0.0124}$$

Because $0.0124 < 0.05$, we reject H_0. At the 5% significance level, the sample provides statistically significant evidence that the population mean difference is not zero.

Suggested allocation: 1 mark each for the parameter, standard error, test statistic, P-value and decision.

2. The 95% confidence interval **[5 marks]**

The margin of error is

$$m_{95} = z^* SE = 1.960(0.12) = 0.2352 \text{ mg/dL}$$

The interval is

$$\bar{x} \pm m_{95} = 0.30 \pm 0.2352 = (0.30 - 0.2352,\ 0.30 + 0.2352) = \boxed{(0.0648,\ 0.5352) \text{ mg/dL}}$$

The null value 0 is outside the 95% interval. Therefore, the corresponding two-sided test rejects $H_0 : \mu = 0$ at $\alpha = 0.05$, which agrees with the P-value decision in Part 1.

Suggested allocation: 1 mark for the margin of error, 2 marks for the endpoints, and 2 marks for explaining the confidence-interval and test connection.

3. Statistical significance, practical importance and nonsampling error **[5 marks]**

The result is statistically significant at $\alpha = 0.05$ because the P-value is 0.0124 and the 95% confidence interval excludes 0. However, the laboratory defines biases with magnitude below 1.0 mg/dL as too small to be practically important. Every value in the 95% interval

$$(0.0648,\ 0.5352)$$

has magnitude below 1.0 mg/dL. The data therefore support a small positive population mean bias, but the estimated bias is not practically important according to the laboratory's stated criterion.

The margin of error describes random sampling variability only. It does not include systematic errors such as calibration drift in the reference instrument, temperature-related measurement bias, inconsistent sample handling or degradation caused by storage. Any one relevant example earns credit.

Suggested allocation: 1 mark for identifying statistical significance, 1 mark for the practical threshold, 2 marks for using the interval to compare the two concepts, and 1 mark for a valid source of systematic error.

4. The 99% confidence interval and the 1% test **[5 marks]**

The 99% margin of error is

$$m_{99} = 2.576(0.12) = 0.30912 \text{ mg/dL}$$

The interval is

$$0.30 \pm 0.30912 = (0.30 - 0.30912,\ 0.30 + 0.30912) = \boxed{(-0.00912,\ 0.60912) \text{ mg/dL}}$$

The 99% interval is wider than the 95% interval because the larger critical value is required to obtain greater long-run coverage. The 99% interval contains 0, so the corresponding two-sided test does not reject H_0 at $\alpha = 0.01$. This agrees with the P-value because

$$0.0124 > 0.01$$

The result is significant at the 5% level but not at the 1% level.

Suggested allocation: 1 mark for the margin of error, 2 marks for the endpoints, 1 mark for the width comparison, and 1 mark for the decision at $\alpha = 0.01$.

5. Sample size for 90% power **[5 marks]**

For the planned upper-tailed test

$$z_{1-\alpha} = 1.645 \qquad \text{and} \qquad z_{\text{power}} = 1.282$$

Substituting the given quantities into the planning formula gives

$$\begin{aligned} n &= \left(\frac{(z_{1-\alpha} + z_{\text{power}})\sigma}{\delta} \right)^2 \\ &= \left(\frac{(1.645 + 1.282)(2.4)}{0.6} \right)^2 \\ &= \left(\frac{(2.927)(2.4)}{0.6} \right)^2 \\ &= (2.927 \times 4)^2 \\ &= (11.708)^2 \\ &\approx 137.077 \end{aligned}$$

The sample size must be a whole number and must provide at least the requested power. Rounding down to 137 could give slightly less than 90% power, so the result must be rounded up:

$$\boxed{n = 138 \text{ canine blood samples}}$$

Suggested allocation: 2 marks for correct substitution, 2 marks for the arithmetic, and 1 mark for rounding up with an appropriate explanation.